

Real-World Dataset Analysis Project

Data Science Assessment – Practical Project

Summary

Analyze a real-world dataset and present meaningful findings using Python, Pandas, NumPy, Matplotlib, Seaborn, and basic statistical techniques.

Dataset Selected

Titanic passenger dataset. The dataset contains information about passengers such as survival status, passenger class, sex, age, number of siblings/spouses, number of parents/children, fare, and port of embarkation. It is suitable for demonstrating data cleaning, exploratory analysis, visualization, and interpretation.

1. Select a Real-World Dataset

Source: Kaggle Titanic dataset / widely used Titanic passenger records. Analytical context: understand passenger survival patterns and examine how variables such as passenger class, sex, age, and fare are associated with survival. Expected outcome: identify important patterns and communicate them through tables and charts.

2. Explore the Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("titanic.csv")
```

```
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
print(df.isnull().sum())
print(df.duplicated().sum())
```

The exploration checks the number of rows and columns, variable names, data types, descriptive statistics, missing values, and duplicate records.

3. Clean & Preprocess Data

```
# Remove duplicate records
df = df.drop_duplicates()

# Fill missing age values with the median
df["Age"] = df["Age"].fillna(df["Age"].median())

# Fill missing embarked values with the mode
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Convert selected columns to appropriate types
df["Survived"] = df["Survived"].astype(int)

print(df.isnull().sum())
```

Potential outliers can be examined using box plots. Extreme values should not automatically be deleted; they should first be checked for whether they represent valid observations.

4. Exploratory Data Analysis

```
# Survival rate
survival_rate = df["Survived"].mean()
print("Overall survival rate:", survival_rate)

# Survival by sex
print(df.groupby("Sex")["Survived"].mean())

# Survival by passenger class
print(df.groupby("Pclass")["Survived"].mean())

# Correlation matrix for numerical variables
print(df.select_dtypes(include=np.number).corr())
```

The analysis examines survival rates, differences between passenger groups, distributions of age and fare, and relationships among numerical variables.

5. Data Visualizations

```
# Survival count
sns.countplot(data=df, x="Survived")
plt.title("Passenger Survival Count")
plt.show()

# Survival by sex
sns.countplot(data=df, x="Sex", hue="Survived")
plt.title("Survival by Sex")
plt.show()

# Age distribution
sns.histplot(data=df, x="Age", bins=20, kde=True)
plt.title("Age Distribution")
plt.show()

# Fare by passenger class
sns.boxplot(data=df, x="Pclass", y="Fare")
plt.title("Fare Distribution by Passenger Class")
plt.show()

# Age vs Fare
sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived")
plt.title("Age vs Fare and Survival")
plt.show()
```

6. Key Findings & Interpretations

- Survival was not evenly distributed across all passenger groups.
- Passenger sex showed a substantial association with survival in the dataset, with female passengers having a higher observed survival rate than male passengers.
- Passenger class was also associated with survival; higher-class passengers had higher observed survival rates than lower-class passengers.
- Age varied considerably among passengers, and age distributions differed across passenger groups.
- Fare values varied by passenger class, with higher-class passengers generally paying higher fares.
- Correlation and group comparisons help identify relationships, but association does not by itself prove causation.

7. Conclusion

The Titanic dataset provides a practical example of a real-world data analysis workflow. The project covered dataset selection, data exploration, cleaning, preprocessing, exploratory data analysis, statistical summaries, visualization, and interpretation. The results demonstrate how Python-based tools can be used to transform raw records into understandable insights.

8. Potential Recommendations

For an analytical report, recommendations should be based on the observed patterns and the purpose of the analysis. In this project, further analysis could include creating age groups, examining survival by combinations of sex and passenger class, applying statistical tests, and building a simple classification model to study survival prediction.

9. Tools Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn.